Beliebers

Brooke Decho, Nimet Yesil, Joshua Okolo, and Ayo Omilola

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Abstract

The design and construction of a remote-controlled robot capable of collecting and placing tennis balls on a vertical surface is a challenging yet significant task. This project is motivated by the need to demonstrate practical applications of robotics in tasks involving precision, adaptability, and efficiency. The ability to navigate inclines and strategically place objects for maximum points showcases the robot's potential in scenarios requiring complex maneuvers and careful planning. The primary challenge addressed by this project is designing a robot that can collect tennis balls, climb inclines of 15° and 30°, and place these balls on a vertical velcro surface at different heights for scoring purposes. The scope of the work involves creating a robot that is not only functional in this specific competition setting but also demonstrates principles applicable to broader robotics applications. To tackle this challenge, our team developed a remote-controlled robot equipped with a servo-motor-driven arm and a 3D-printed bucket. The arm, made of high-tensile-strength acrylic, was designed to reach the highest scoring zone on the velcro surface. The bucket, with a capacity for two balls, ensured efficient scoring. We addressed mobility and power by employing two DC motors and a belt-pulley system for four-wheel drive, along with a 1:4 gear ratio in the drivetrain for climbing inclines and a 1:3 gear ratio in the arm for lifting. The design process involved both prototype construction and iterative testing to refine the robot's capabilities. During the competition, the robot successfully scored by placing a ball on the velcro surface. However, it encountered unforeseen challenges, such as getting stuck on the ramp corners, which limited its performance. Despite this, the robot demonstrated its core functionalities, including climbing, ball collection, and placement, albeit getting stuck and time constraints affecting its overall scoring potential. The results indicate that while the robot achieved its primary objectives, there is room for improvement in handling unforeseen environmental challenges. These findings are specific to the robot's current design but offer valuable insights for future iterations and similar robotics projects. The experience underscores the importance of thorough testing in varied conditions to enhance a robot's adaptability and reliability in real-world applications.

Concept Development:

In the development of our remote-controlled robot, our team faced several critical constraints that shaped our design and fabrication approach. Manufacturing-wise, we were required to employ a diverse set of fabrication methods, including milling, laser cutting, turning, 3D printing, drilling, and molding, with the caveat that the methods used in assembling our gearbox did not contribute towards this requirement. The project was also bound by a stringent time frame, with the design and construction process spanning from September 2023 to December 4th of the same year. A pivotal design constraint was the necessity for the robot to fit within the confines of a "Box of Justice," which had a limiting volume of 12x12x12 inches. In terms of motorization, our design was restricted to using 2-3 motors sourced from an electric screwdriver, supplemented by up to 4 additional actuators or servos, not exceeding a total of 6 actuators. Moreover, these motors were not permitted to draw more than 9 amperes of current. Materially, our resources were confined to a maximum of 15 cubic inches of 3D printing material, alongside 2 sheets each of 12”x24”x1/4” and 12”x24”x1/8” acrylic, with a strict prohibition on using materials from outside the lab. These constraints significantly influenced the overall design, construction, and functionality of our robot.

The factors our team considered when assembling our robot included the efficiency of collecting the balls, mobility, design and control simplicity, weight and scoring capabilities. We placed more weight on mobility and the scoring capabilities of our robot, as these are the factors that would make our robot most successful in the competition. We rated our scoring capability as 5/5 because we can carry two balls at a time and place them on the third zone of the velcro. We rated our design and control simplicity as 3.5/5 because we only need one servo to lift the balls up, however it will still be a challenge to collect two balls at the same time. Our robot design has a low center of gravity, which makes it easier for the robot to climb the ramps, and contributes to the mobility score of 4/5. We gave the weight of our robot a score of 1.75/5 because we will be lifting two balls at once, which increases the weight of the bucket. Also the bucket has to be larger. The collection efficiency of our design scored 0.5/5, because it will be challenging to ensure that the balls go inside the bucket. We decided to make an angle design on the bucket to keep the balls in the bucket.

Alternative Solutions:

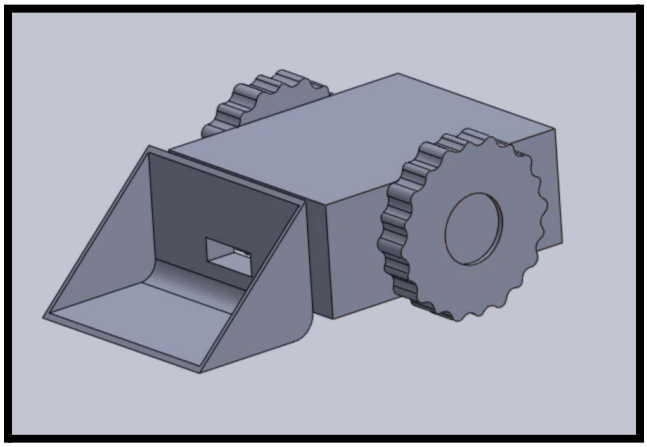


Figure 1: Bulldozer design

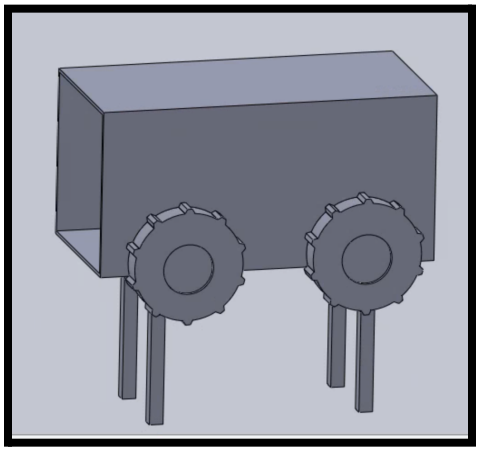


Figure 2: Elevator design

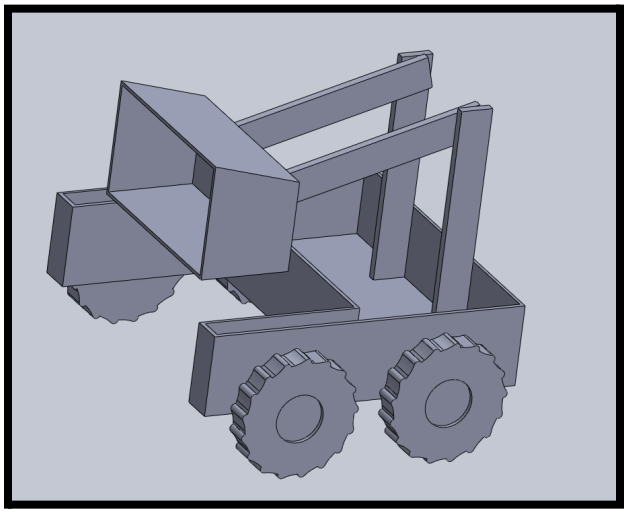


Figure 3: Bucket Lever

The first design was inspired by a bulldozer, and scoops a ball and places it on the velcro. A drawback of this design is that the robot can't place the balls on the third zone of the velcro, which would limit the amount of points our team can get. The second design is an elevator design, where the robot starts flat, but then has spring stilts that pop out raising the balls up and placing them on the velcro. We thought this design allowed us to collect more balls, however it would be too difficult to spontaneously lift the entire robot due to its large mass. In the end we went with the third design because of its simplicity, mobility and scoring capability.

Table 1: Pugh Matrix used to select our design

Criteria	Baseline	Weight	Bulldozer	Elevator	Our Design
Collection Efficiency	0	2	2	-2	0.5
Mobility	0	6	1	1	4
Design + Control Simplicity	0	4	4	1	3.5
Weight	0	2	2	1	1.75
Scoring Capabilities	0	6	1	3	5
Total	0	20	10	4	14.75

Analysis

Drivetrain Calculations:

First we calculated the required wheel torque required to go up the incline. We then compared this value with the maximum tractive torque. The required wheel torque was greater than the tractive torque, which means we needed a gear ratio.

$$\tau_w = \frac{mgr}{n} sin\theta + \frac{v}{gt} = \frac{3.110kg*9.81m/s^2*0.0508m}{2}(sin(30) + \frac{0.15\,m/s}{9.81m/s^2}) = 0.40Nm$$

$\tau_{max} = \mu m_w g cos\theta r = 0.8 * 0.44kg * 9.81m/s^2 = 0.15Nm < 0.40Nm$

Required Gear ratio:

$\tau_m = 0.308Nm,\ GR = \frac{(1.5)(0.40Nm)}{0.308Nm} = 1.948 < 3$

Our gearbox provides a 1:3 gear ratio, so the robot will be able to climb the incline.

We conducted a gear train demonstration to ensure that our robot can climb up the incline. We also tested our belt and pulley system to make sure our four wheel drive works well.

Maximum torque required to lift and hold the balls:

We calculated the torque required to lift two balls, and compared it to the operating torque of the servo motor.

$arm\ length\ (r) = 8'' = 0.2032m \qquad f = 144g = 1.412N$

$\tau = r \times f = 0.2032m * 1.412N = 0.291Nm = \tau_{required}$

$\tau_{stall}\ at\ 6V = 0.67Nm, \tau_{operating} = \frac{0.67Nm}{2} = 0.335Nm$

The servo provides 0.335 Nm of torque and our arm requires 0.291 Nm. We will still include a gear ratio of 4, to ensure that the arm has enough torque to lift the balls. We tested the servo motor to see if it can rotate the arm.

Loading Conditions:

We calculated the maximum bending moment on our arm as -48.84 Ncm. We used this value to calculate the maximum stress experienced by our arm. Figure 4 below shows the shear force and maximum bending moment experienced by the arm, as well as a calculation of the maximum stress:

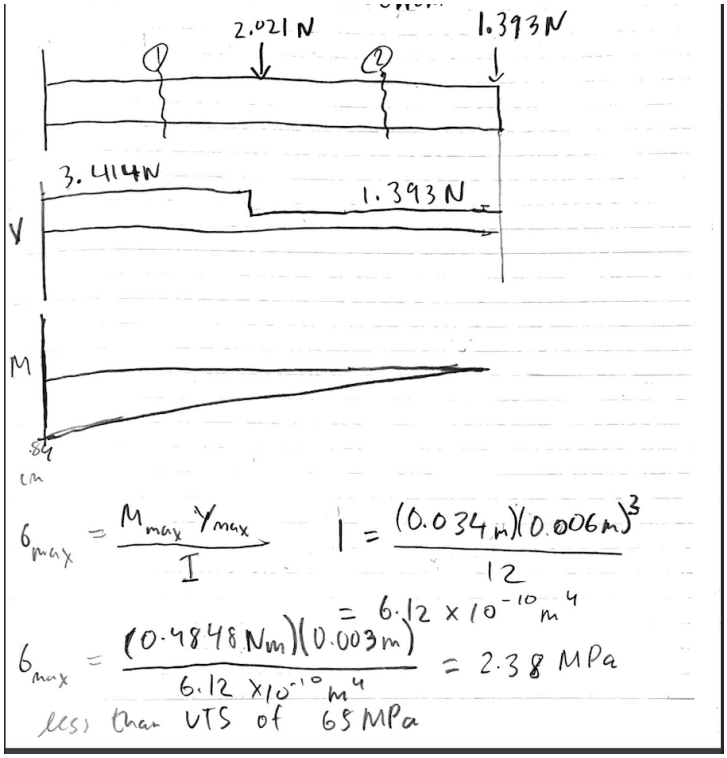


Figure 4: Shear Force, Bending Moment graphs and Max Stress

As shown in figure 4, the maximum stress experienced by our beam is 2.38 MPa with an empty bucket. This is much less than acrylic's ultimate tensile strength of 65 MPa, so we believe we can carry at least two balls.

Final Solution

Our final design solution was able to meet the primary requirements of the competition, including collecting balls, driving up the incline and successfully placing a ball on the velcro. Our robot experienced difficulties in its drivetrain during the competition, as the robot was not driving as smoothly as we would have wanted it to. This limited the amount of points we could score, and reflected our position in the tournament.

Our robot satisfied the criteria outlined in the ES 51 Turf Wars rule book. We designed a fully functioning robot, using only the materials provided to us, as well as the six different machining techniques we have learned in class. Our fully assembled robot fit in the "Box of Justice", and had enough torque to climb the ramps and lift balls, using only 3 actuators.

The drivetrain of our robot made it difficult to go around sharp corners in the competition. This limited the speed at which we could place balls on the velcro. The drivetrain design could be improved by making sure the shafts are arranged in a way that allows the wheels to rotate freely. Additionally, we coil have designed a smaller robot base, to help us get around the corners of the arena more quickly. Another area

of improvement for our robot is to modify the bucket design. Instead of focusing on collecting multiple balls at once, we could have focused on collecting and placing one ball at a time. This simplifies the problem, and allows us to improve the speed and efficiency of the robot. This would also reduce the size of the robot, and help it get around corners much faster.

Final Design Specifications

Table 2: Robot Design Specifications

SPECIFICATION	VALUE	NOTES
Mass	2.755 kg	Mass of robot without batteries.
Robot Dimensions	11.75” L x 11.5”W x 11.5”H	Height when the arm is down
Undercarriage Clearance	0.25”	The back of the bucket has a clearance of 0.25”, the front touches the ground to collect the balls.
Turning Radius	12 in	Measured with a ruler
Drivetrain Gear Ratio	3:1	48 teeth to 16 teeth gear ratio
Arm Gear ratio	4:1	64 teeth to 16 teeth gear ratio.
Wheel Diameter	4’’	Measured with a caliper
Bucket Dimensions	5.875” L x 3.75”W x 5”H	Measured with a caliper.
Arm length	17.75”	Max height from robot base to base of bucket

Appendix:

Table 3: Bill of Materials:

Item	PART NAME	DESCRIPTION	QTY	MATERIAL/

no.				CONSTRUCTION
1	Base Plate	¼’’ laser cut robot chassis	1	Acrylic, laser cut
2	Planetary Gearbox	Planetary gearbox from electric screwdriver	2	Assorted
3	Motor	LS 380S/385S from electric screwdriver	2	Assorted
4	Wheel with hex	Robot drive wheels	4	Silicone Coating, molding
5	Motor Mount	Holds the motors	2	PLA (Add amount used), 3D printing
6	Delrin support	Delrin shaft support	2	Delrin, CNC mill, drill press
7	Aluminum shaft support	Aluminum shaft support for the rear wheels	2	6061 Aluminum, Drill press, horizontal and vertical bandsaw
8	Laser cut front plate	⅛ “Laser cut front plate for motors.	2	Acrylic, laser cut
9	97431A300	E-Clip 0.25" ID	2	1060-1090 spring steel
10	95606A120	Nylon Washer 0.25" ID	24	Nylon
11	Washer Screwdriver Small	Washer 0.41" ID, 0.54" OD from Screwdriver	2	Stainless Steel
12	Washer Screwdriver Large	Washer 0.39" ID, 0.77" OD from Screwdriver	2	Stainless Steel
13	Metal Circle	Metal Circle with 6 notches from Screwdriver	2	1215 Carbon Steel
14	Hex Drive Shaft	Hex Drive Shaft from Screwdriver	2	6061 Aluminum
15	92010A022	M2.5 x 12 mm Flat Head Screw	11	18-8 Stainless Steel
16	91772A116	Pan Head Machine Screw 4-40 Thread 1.25"	8	18-8 Stainless Steel
17	94639A662	Unthreaded Spacers, 3/16" OD, 11/16" Length	8	Nylon
18	90126A505 Washer	Washer for Number 4	10	Stainless Steel

		Screw Size, 0.125" ID, 0.312" OD		
19	91772A110	Pan Head Machine Screw, 4-40 Thread, 0.5" Length	8	18-8 Stainless Steel
20	98541A118	External Retaining Ring for 10mm Shaft Diameter	2	15-7 PH stainless steel
21	91772A108	Pan Head Machine Screw, 4-40 Thread, 3/8" Length	4	18-8 stainless steel
22	Nylon Bushing	Nylon Flange Bushing, ID 1/4" OD 3/8"	4	Nylon, Broached
23	A 1M 2-TA32016	Acetal Plastic Gear, 16 Teeth, 32 Pitch	2	Acetal Plastic, Broached
24	A 1M 2-TA32048	Acetal Plastic Gear, 48 Teeth, 32 Pitch	2	Acetal Plastic, Broached
25	92373A113	Slotted Spring Pin, 1/16"	2	18-8 Stainless Steel
26	91099A169	Flat Head Phillips Machine Screw, 4-40 Thread, 5/8" Length	4	18-8 Stainless Steel
27	Analog Servo	Servo motor that rotates the arm	1	Assorted
28	Laser Cut Arm	¼ ‘’ laser cut arm	2	Acrylic, laser cut
29	Battery	Battery that powers the robot.	1	Assorted
30	Long aluminum L bracket wheel support	Aluminum Shaft Support for outside of wheels	2	6061 Aluminum, Horizontal & Vertical band saw, & Drill press
31	Bucket	Bucket that carries the balls.	1	PLA (Add amount used), 3D printing
32	A 1M 2-TA32064	Acetal Plastic Gear, 64 Teeth, 32 Pitch	1	Acetal Plastic, Broached
33	Arm rod	Hex shaft rods that ensure that the two ends of the arm are stable and rotate at the same time.	3	6061 Aluminum
34	1679K117 Timing Belt	Timing Belt, 200" pitch, 18"	2	Urethane

		Outer circle		
35	6495k711 Timing Belt Pulley	Timing belt for our four wheel drive	4	PLA 3D printed
36	Milled Delrin	Milled Delrin to support the base of our arms.	2	Delrin, CNC mill, Drill Press
37	Shaft Coupler	Shaft coupler for rear wheels	2	Stainless Steel
38	Aluminum L bracket	Aluminum L brackets to attach the front wheels to the base	2	6061 Aluminum, Horizontal & Vertical band saw
39	91772A121	Flat Head Phillips Machine Screw, 4-40 Thread, 2" Length	1	18-8 Stainless Steel
40	90126A505 Washer	Washer for Number 4 Screw Size, 0.125" ID, 0.312" OD	6	Stainless Steel

<u>3D print material used:</u>

Volume of bucket = $mass/density = 150g/20.48383g/in^3 = 7.323in^3$

Volume of timing belt = $mass/density = 4.535924g/20.48383g/in^3 * 4 = 0.886in^3$

View drawings of 3 non standard parts with dimensions:

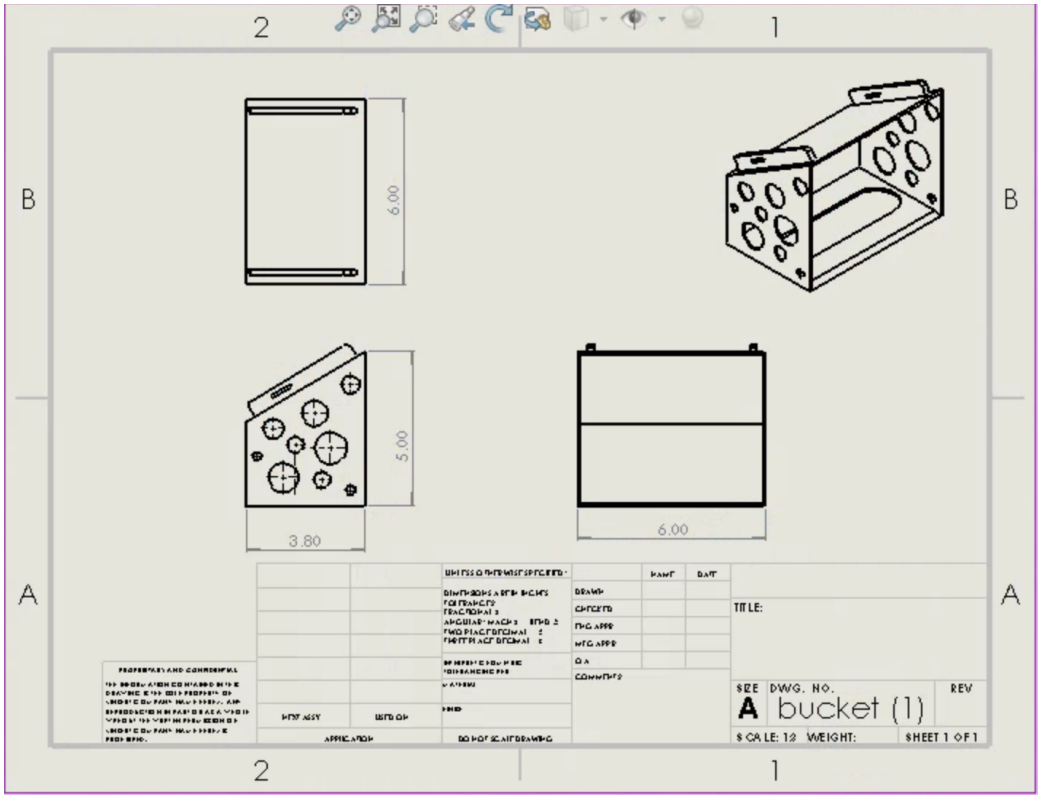


Figure 5: Drawing of bucket

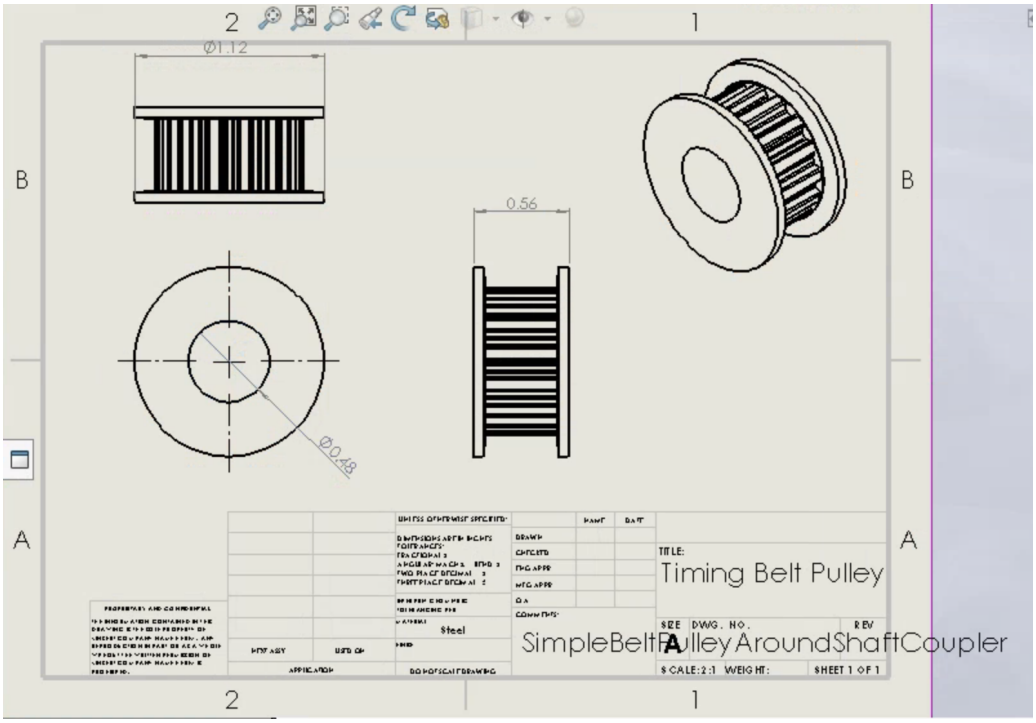


Figure 6: Drawing of Simple Belt Pulley Around Shaft Coupler

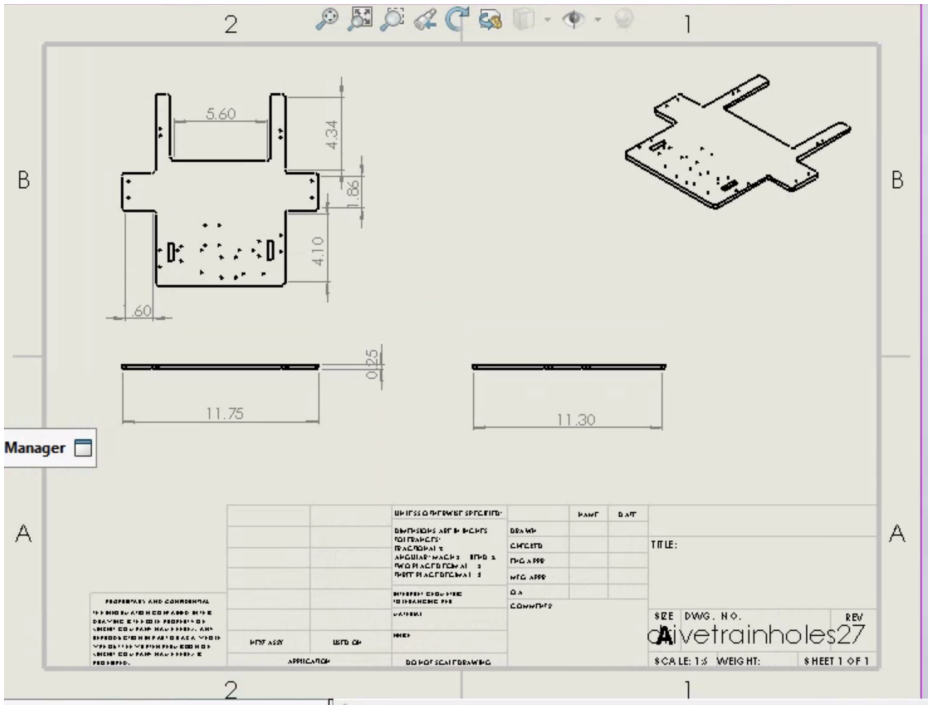


Figure 7: Drawing of Drivetrain with holes

Assembly Drawing of Final Robot:

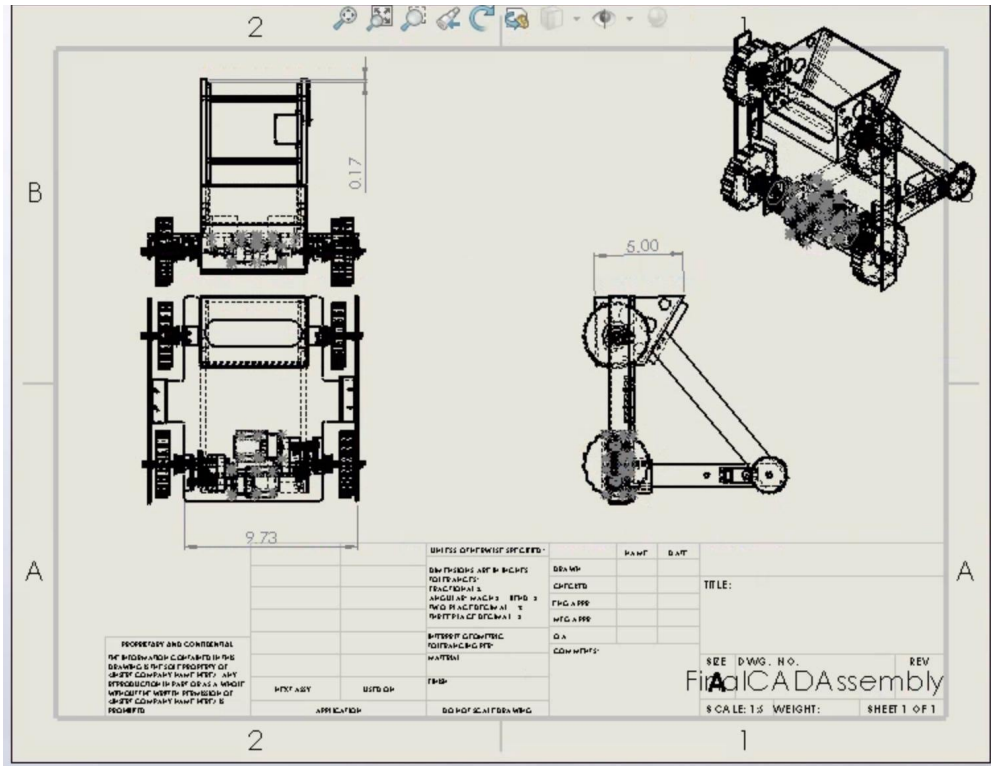


Figure 8: Assembly Drawing of Robot

Exploded View of Robot:

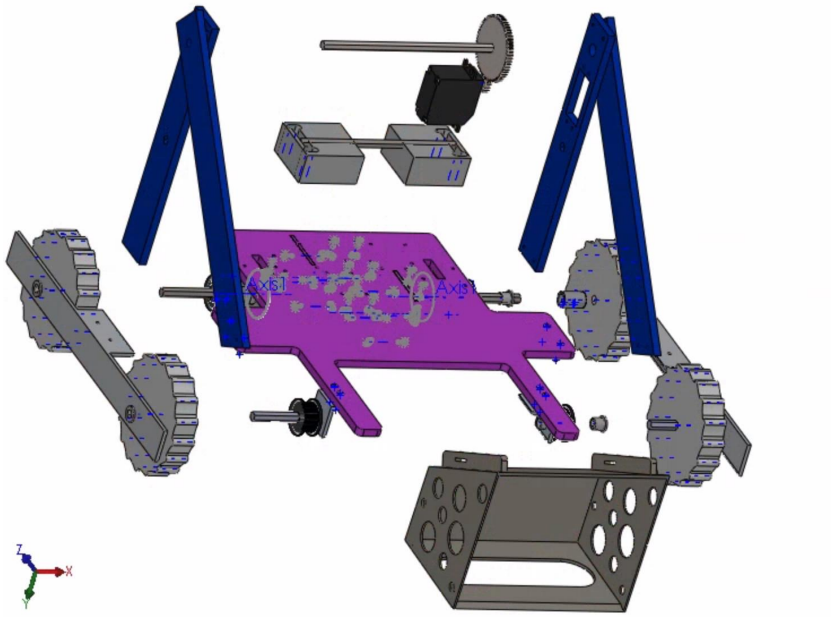
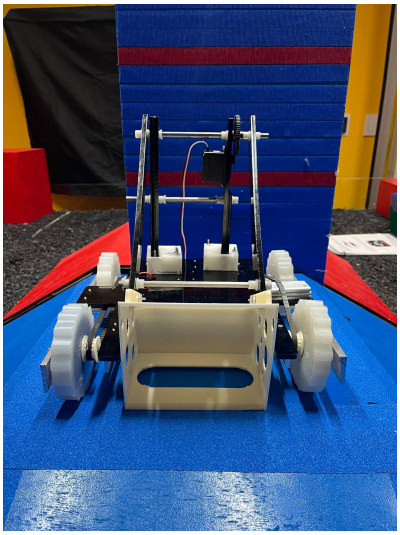


Figure 9: Exploded View of Robot

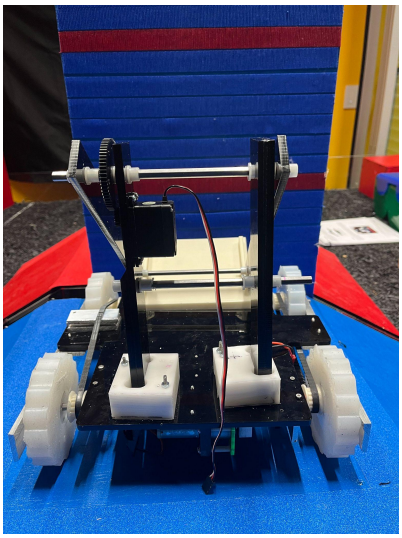
Pictures of final robot:



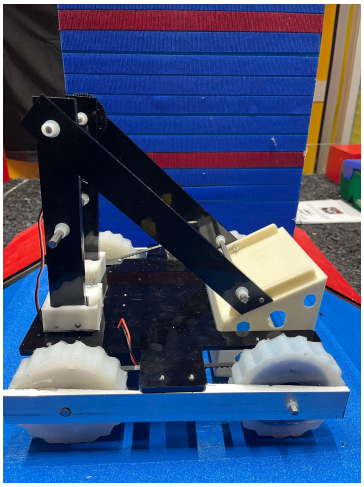
Front View



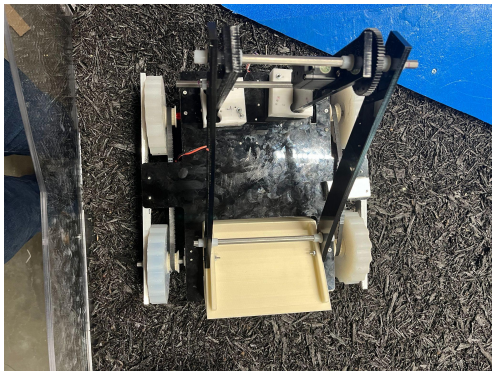
Right View



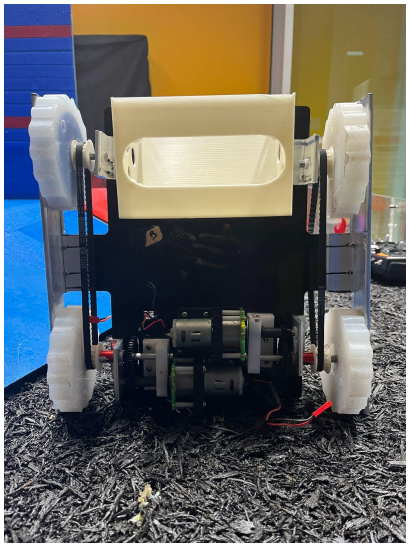
Back View



Left View



Top View



Bottom View